COMPUTER NETWORK

UNIT- I

TCP/IP Protocol Suite

① Layered Architecture :-

In data communication and networking, a protocol defines the rules that both the sender and receiver and all other intermediate devices need to follow to be able to communicate effectively.

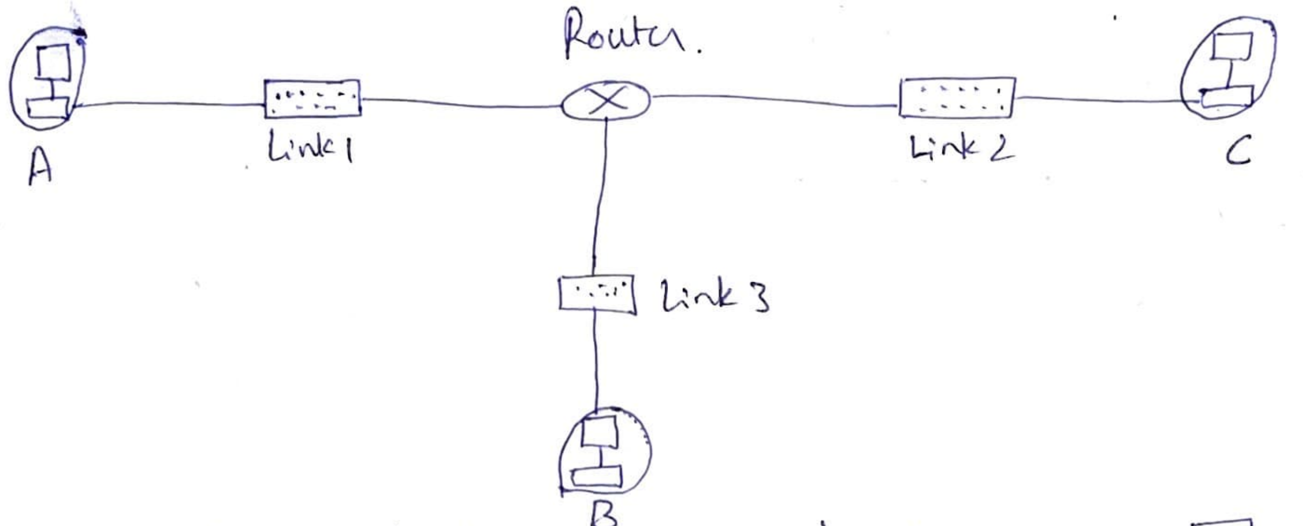
A set of protocols organized in diff layers known as TCP/IP protocol suite which is used in internet in current days.

It is a hierarchical protocol madeup of interactive modules, each of which provides particular functionality.

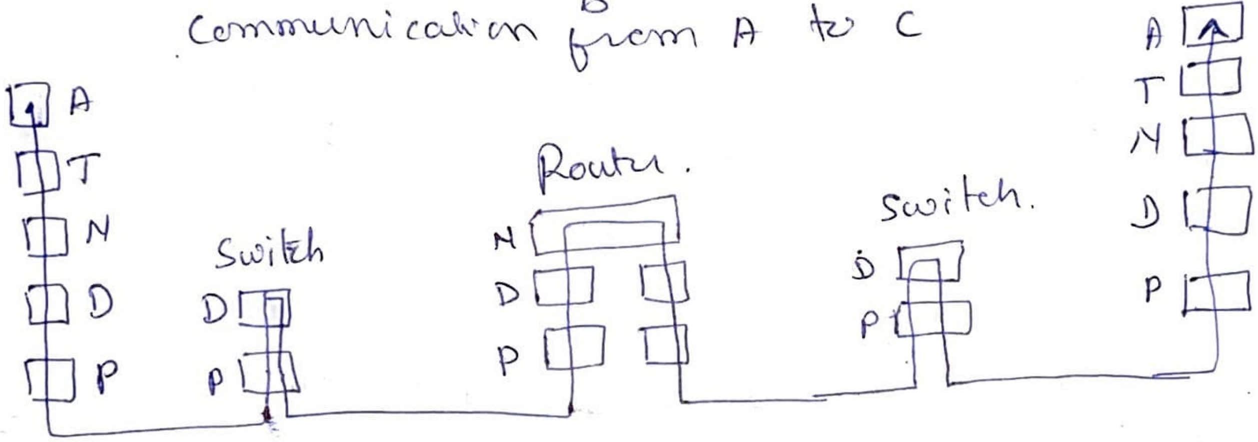
The term 'hierarchy' means, each upper level protocol is supported by the services provided by one/more lower level protocols.

① original layer		currently used layer	
Application	⟷	Application	L5
Transport	⟷	Transport	L4
Internet	⟷	Network	L3
Network interface	⟷	Data link	L2
Hardware device	⟷	Physical	L1

Let us consider the following scenario to understand the involvement of layers of TCP/IP in communication



Communication from A to C

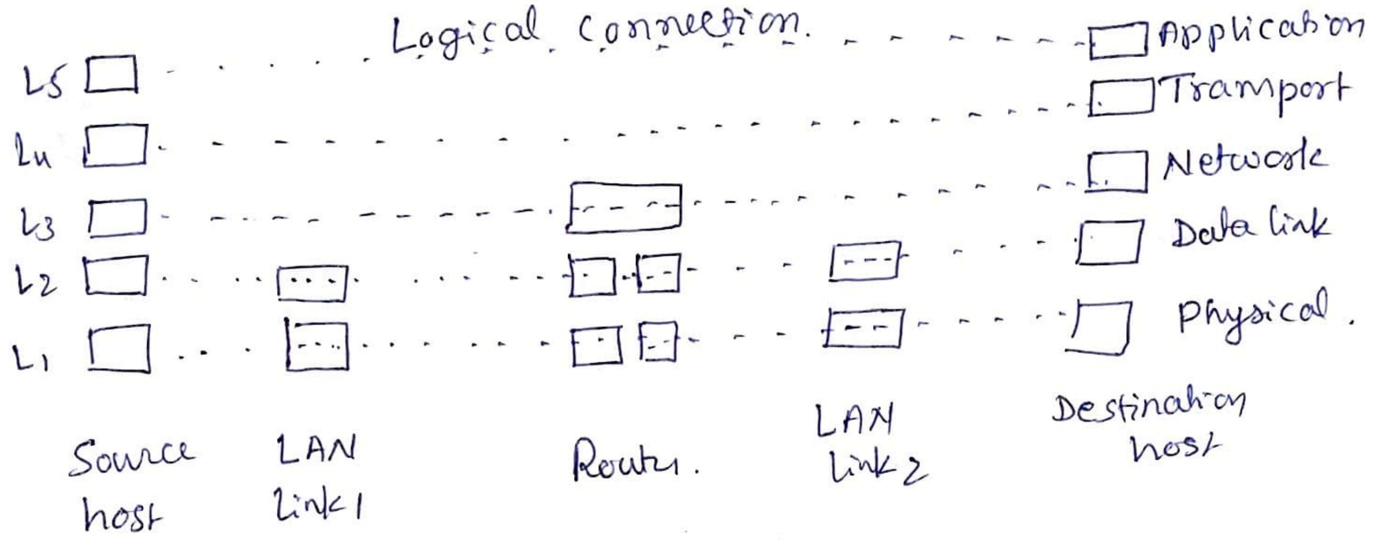


The host A needs to create a msg in the application layer and send it down the layers so that it is physically sent to the destination.

The host C needs to receive the communication at the Physical layer and then deliver it through the other layers to reach the application layer.

Layers in TCP/IP protocol suite

For better understanding of functionalities of each layer we need to think about the logical connections between layers. The dotted lines in the fig represents the logical connection



From fig it can be seen that the duties of Application, transport, and n/w layers is **end-to-end**, while the duties of the data-link and physical is **hop-to-hop** in which a hop in a host or router.

The domain of duty of top three layers is internet
The domain of duty of bottom two layers is link.

The data unit (packet) should not be changed at both switch & router in top-three layers.
The packet can be changed in bottom layers of only at routers.

Description of each layer.

① Physical Layer.

It is responsible for carrying individual bits in a frame across the link. Two devices are connected by a transmission medium (cable or air). But, this media cannot carry bits instead it carries electrical or optical signals, so therefore bits are to be converted into signals. (Digital to Analog)

② Data link layer

we know that, Internet is made up of several links (LANs and WANs) connected by routers, which is responsible for choosing the best link. When a link is decided by the router, it is the responsibility of link layer to carry to next hop.

Responsibilities of Data link layer.

① It works at hop-to-hop level. (hop-to-hop delivery)

② DLL is sufficient to communicate bet 2 nodes within a specific n/w.

③ Flow control

protocols : Stop and wait
Go Back N : GBN.

DLL will control the msg flow at each node

④ Error control : Erly detection of error.

methods : cyclic redundancy check
Checksum.
parity bit set

⑤ Access control : Improves the accessibility of n/w.

protocol : CSMA/CD : Carrier Sense Multiple Access / collision Detection

⑥ Framing - Data packets are converted into fixed sized frames which increases reliability.

Header	Packet	Trail

DLL divides the stream of bits received from n/w layer into manageable data units called frames.

③ Network Layer

① Responsible for source-to-destination delivery of a packet, possible across multiple n/w (links)

② It uses logical addressing (IP)

③ Routers in the path are responsible for choosing the best path to reach destination.

④ IP - It defines the format for packet called datagram

⑤ IP also defines the format and the Structure of the addr used in this layer.

⑥ Responsible for routing - forward the pkt at each router.

⑦ IP is a connectionless protocol that provides no flow control, no error control and no congestion control.

⑧ This layer also has some auxillary protocols that help IP in its delivery & routing tasks.

ⓐ ICMP - Internet Control Msg protocol help IP in routing

ⓑ IGMP - Internet Group Management Protocol help in Multitasking

ⓒ DHCP - Dynamic Host Configuration Protocol helps to get n/w layer addr for host

ⓓ ARP - Address Resolution Protocol help to find the link-layer addr of a host/router when its n/w layer addr is given

(4) Tramsport Layer

(1) Responsible for end-to-end communication
(2) Segmentation is done here to get the msg/pkt in the form of segment/user datagram.

(3) Sends the segment through logical connection, to the tramsport layer at the destination host.

(4) Transmission Control Protocol (TCP) is connection oriented protocol

(5) TCP establishes a logical connection bet source and destination host before transferring the data.

(6) Responsible for error, flow, and congestion control.
(7) UDP - user Datagram Protocol - connectionless
ie user datagram can be transmitted without first creating a logical-connection

(8) UDP will not provide error, flow, or congestion control
(9) UDP is simple protocol, hence lower overhead, therefore Appl^n layer prefer UDP in case of small msg transfer.

(10) SCTP (Stream Control Transmission Protocol) is designed to respond to new appl^n that are emerging in the multimedia.

⑤ Application Layer.

① Comⁿ at AL is bet 2 processes (2 programs) is process-to-process comⁿ.

② HTTP (HyperText Transfer Protocol) used for accessing www.

③ SMTP (Simple Mail Transfer Protocol) used in e-mail (electronic-mail) service

④ FTP (File Transfer Protocol) used for transferring files bet hosts to host.

⑤ TELNET (TErminaL NETwork) and SSH (Secure SHell) are used for accessing the site remotely.

⑥ SNMP (Simple Network Management Protocol) used by administrator to manage the Internet at local and global levels.

⑦ DNS (Domain Name System) - used to find the n/w layer address

⑧ IGMP (Internet Group Management Protocol) used to collect membership in group

Encapsulation and Decapsulation

Encapsulation

① TL adds its header (4) to the pkt which includes identifiers of the source and destination applⁿ prog & also infⁿ related to flow, error & congestion control.

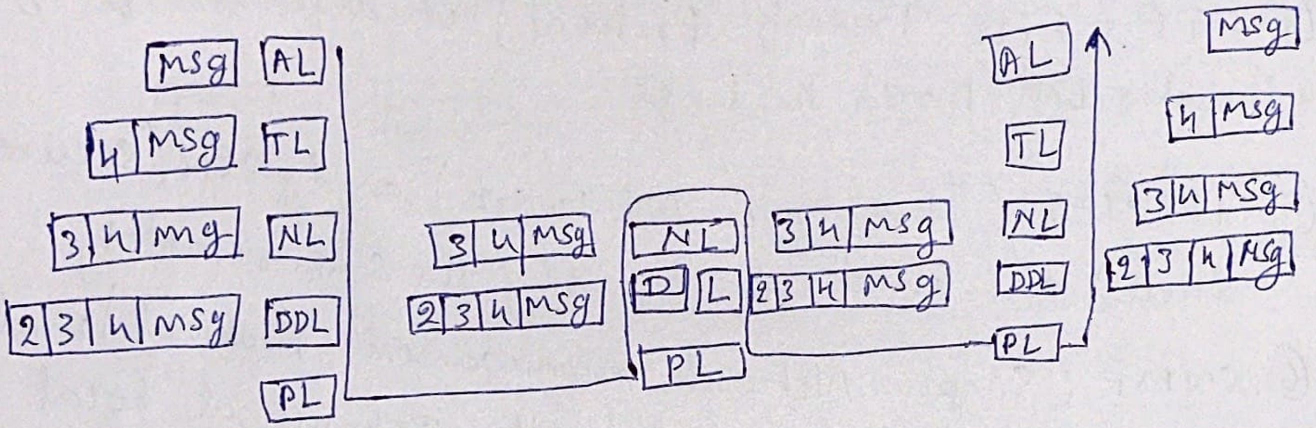
② NL adds its header (3) to ~~the~~ the pkt which includes address of source & destination nodes

③ DL adds its header (2) to datagram which includes link layer address.

Decapsulation

Removes the payload & transfer or deliver it to the next-higher layer

The process of encapsulation & decapsulation at each layer is given in the below diagram.



Addressing

Pkt Names	Layers	Address
msg	Appl^n Layer	Names
segment/user datagram	Transport layer	Port Numbers
Datagram	Network layer	Logical Address
Frame	Data link layer	Link-layer Address
stream of bits.	Physical layer	—

Multiplexing

A protocol at a layer can encapsulate a pkt from several next next-higher layer protocol (one at a time)

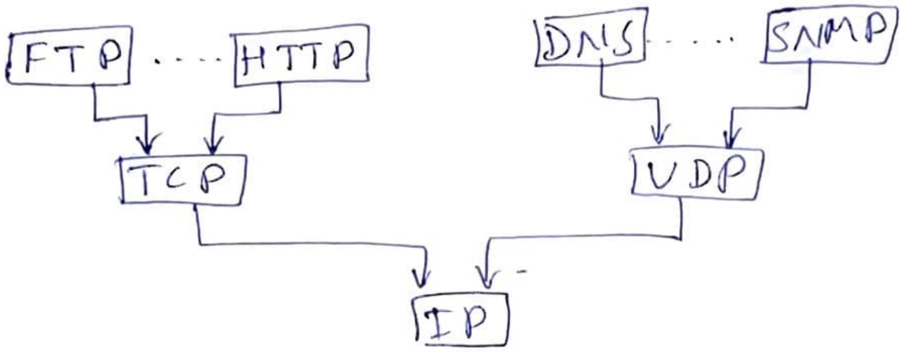
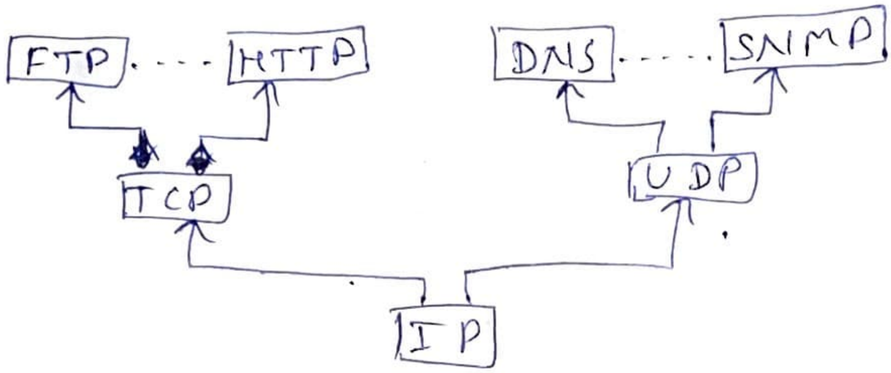


Diagram of multiplexing process at source

Demultiplexing

A protocol can decapsulate and deliver a pkt to several next-higher layer protocol (one at a time)



Demultiplexing process at destination.

3.5 Data Rate Limits

An important consideration in data comⁿ is how fast we can send data in bits/sec. Data rate depends on 3 factors.

1. The bandwidth available
2. The level of the signals used
3. The quality of the channel (level of noise)

Two theoretical formulas were developed to compute data rate
1. Nyquist formula for noiseless channel
2. Shannon formula for noisy channel

1. Noiseless Channel : Nyquist Bit Rate

This formula defines the theoretical max bit rate as

$$\boxed{\text{BitRate} = 2 * \text{Bandwidth} * \log_2 L}$$

where
Bandwidth is the bandwidth of a channel
L is the no of signal levels used to represent data
BitRate is bits/second

According to the formula increase in L will increase the BitRate, but may reduce reliability of system

Ex: ① Consider a noiseless channel with a bw of 3000 Hz transmitting a signal with 2 signal level then BR = ?

Soln :- BitRate $= 2 * 3000 * \log_2 2$
$= 6000$ bps.

② We need to send 265 kbps over a noiseless channel with a bw of 20 kHz. then how many signals do we need?

Soln :-

$265000 = 2 * 20000 * \log_2 L$

$\log_2 L = \frac{265000}{2 * 20000} = \frac{265}{40} = 6.625$

$L = 2^{6.625} = 98.7$ levels

⑨ Noisy channel : Shannon Capacity
In 1944 Claude Shannon introduced a formula called Shannon capacity to determine the theoretical highest data rate for noisy channel.

$$\boxed{\text{Capacity} = \text{bandwidth} * \log_2 (1 + SNR)}$$

where SNR is signal-to-Noise ratio

4.1. Digital-to-Digital Conversion

Line Coding

① It is a process of converting digital data to digital signal.

② Generally the data will be in the form of text, audio, video, graphical image, numbers, etc

③ This digital data is stored on a computer memory in the form of sequence of bits

④ Line coding converst a sequence of bits into digital signal.

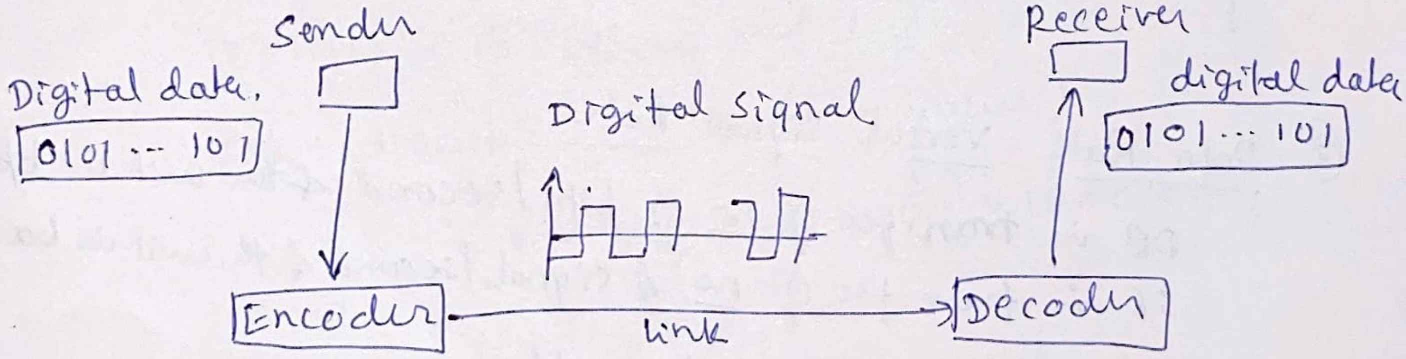
I/P — Line coder — O/P

Digital data → $\boxed{\text{Encoding}}$ → Digital signal.

Digital signal → $\boxed{\text{Decoder}}$ → Digital data

Encoding is done at sender.

Decoding is done at receiver



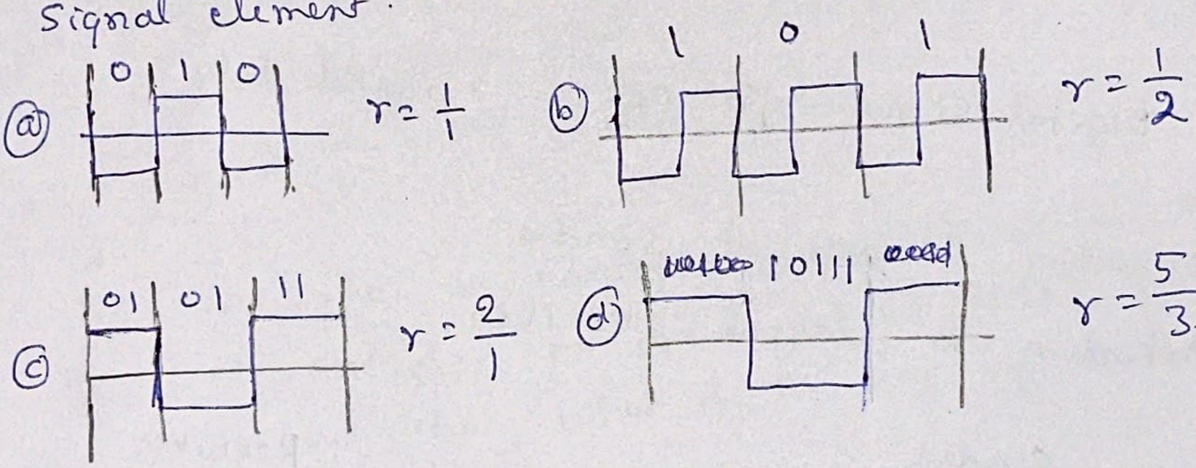
characteristics

① signal element versus Data Element

* The goal is to send the data element
* Data element is the smallest entity that can represent piece of infn is bit
* Signal ele carries data ele
* Signal ele is a shortest unit (timewise) of a digital signal
* Data ele is being carried while signal ele is carrier

Let r be the ratio representing no of data ele carried by each signal ele. ie $r = \frac{\text{no of Data elements}}{\text{no of signal elements}}$

Following diagram represents the data ele & corresponding signal element.



② Data Rate versus Signal Rate

DR is transfer of no of bits/second & the unit is bps

SR is transfer of no of signals/second & the unit is baud

Data Rate is also called bit rate

signal Rate is also called pulse rate/modulation rate/baud rate

main goal in data comn is to increase bit rate and decrease signal rate.

* bit rate ↑ will ↑ the transmission speed
* Signal rate ↓ will ↓ the bandwidth requirement.
* The relationship bet DR (N) & SR (S) is given as

$$S = N / r.$$

* The relation mainly depends on data pattern (r)
* To device the proper formula for the relationship we need to define 3 diff cases
(a) worst (b) Best (c) Average
* worst case is when there is high SR while best case is when the SR is minimum. However, we are interested in avg case which can be reformulated as

$$S_{avg} = c * N * \frac{1}{r} \text{ baud.}$$

where N is data rate in bps.
c is case factor varies for each case
r is ratio (DE/SE)

Ex:- A signal is carrying data in which one data ele encoded on one signal ele (r=1). If the bit rate is 100 kbps, what is the avg value of the baud rate if c is bet 0 and 1 ?

Soln :- Assume $c = 1/2$ (avg value)

$$S = c * N * \frac{1}{r} = \frac{1}{\cancel{2}} * \overset{50}{\cancel{100}} * 1000 * \frac{1}{1}$$

$$= 50000$$

$$= 50 \text{ kbaud}$$

③ Band width

Baud rate deterministic the required bandwidth for digital signal

$$B_{min} = c * N * \frac{1}{2}$$

$$N_{max} = B * \cancel{r} * \frac{1}{c}$$

④ Baseline wandering

In decoding a digital signal, the receiver calculate a running avg of received signal power and this avg is called baseline.

Then, the further incoming signal power is evaluated against this baseline to determine data ele.

The long strings of 0's or 1's may cause drift in baseline i.e. baseline wandering & make it difficult for receiver to decode correctly

⑤ DC component

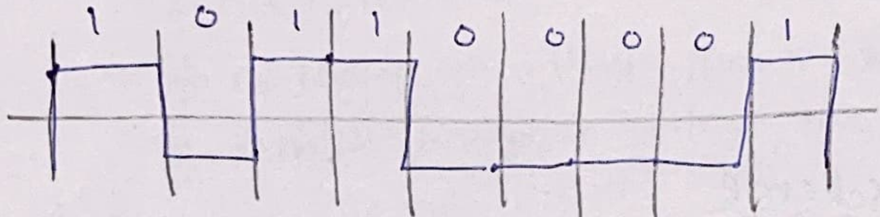
When voltage levels in a digital signal is consistant or constant for a while, the spectrum creates very low frequencies. These frequencies around zero are called DC [Direct current] component.

This causes problem for system that cannot pass low frequencies or the system that uses electrical coupling

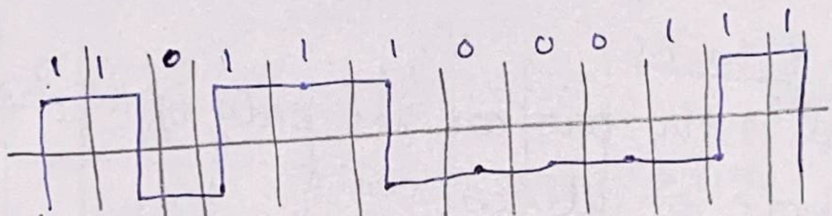
DC component means 0|1 parity that can cause Baseline wandering. Ex:- Telephone lines cannot pass frequencies below 200 Hz. For such systems we need a scheme that don't have DCC

⑥ Self-synchronization

To correctly interpret the msg the receiver's bit intervals must correspond exactly to the sender's bit interval. If the receiver clock is faster/slower, than sender, then the bit intervals are not matched & the msg may be misinterpreted by receiver



Sent signal



Received signal. The receiver has shorter bit duration

A self-synchronizing DS (Digital signal) includes timing info^n in the data being transmitted.

Ex:- In a digital transmission, The receiver clock is 0.1% faster than the sender clock. How many extra bits/sec does the receiver receive if the data rate is 1 kbps? How many, if data rate is 1 Mbps?

soln:- At 1 Kbps the receiver receives 1001 bps instead of 1000 bps.

At 1 Mbps the receiver receive 1001000 bps instead of 1000000 bps.

⑦ Built-in-error detection

It is desirable to have a built-in-error detection capability in the generated code to detect some/all error that occurred during transmission.

⑧ Immunity to Noise - and Interferences

The generated code must be immune to noise and other interference.

⑨ Complexity

complex Schemes are more costly to implement, then it is expected to have coding scheme which is less complex.

Line coding scheme

① Unipolar Scheme

All Signal levels are on one side of the time axis is either above or below axis

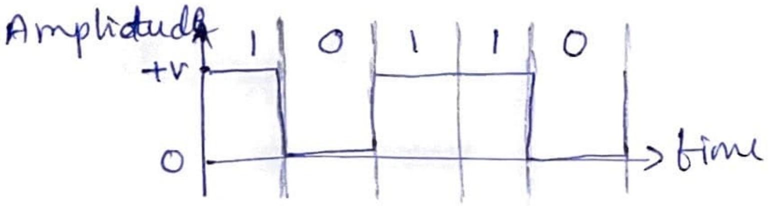
NRZ [Non-Return-to-Zero]

Traditionally unipolar scheme was designed as a NRZ scheme

+V voltage define the bit 1

0 voltage define the bit 0.

The name NRZ is because the signal does not return to zero at the middle of the bit.



② Polar scheme

voltage are on both side of the time axis.

voltage level for 1 is +V

Voltage level for 0 is -V.

NRZ polar : There are 2 version of polar NRZ

ⓐ NRZ - L (Level)

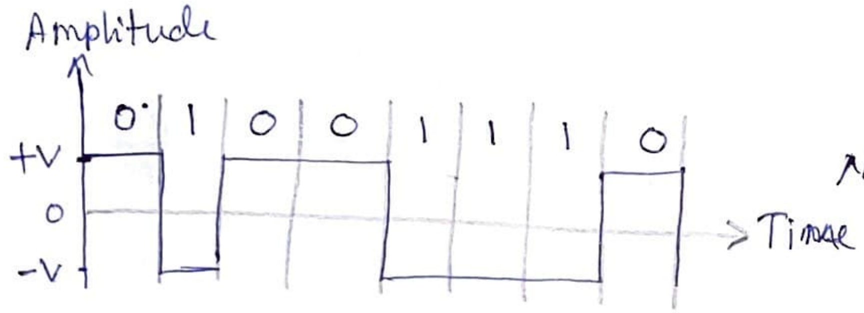
Voltage level determines the bit value

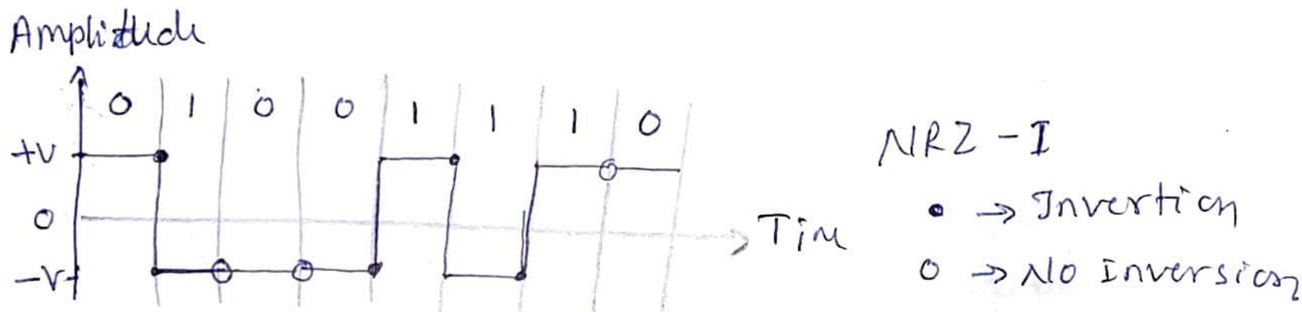
ⓑ NRZ - I (Inversion)

Change or no change in voltage determine the bit value.

If no change in voltage then the bit value is 0

If there is change in voltage, the bit value is 1.





NRZ - L : problem with both continuous zero's and one's

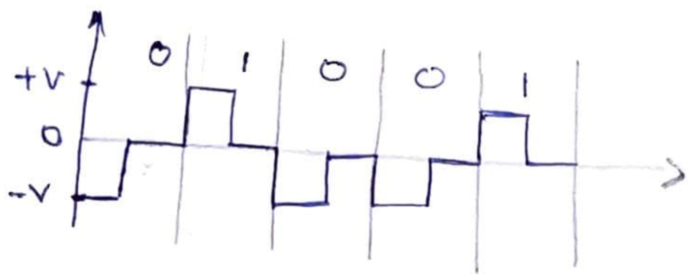
NRZ - I : problem with only continuous zero's.

The continuous 1's or 0's leads to baseline wandering and DC component occurrence

RZ (Return-to-Zero) Polar

As we know, main problem occur when sender and receiver clocks are non-Synchronous

The receiver doesn't know when the bit started and ended. Solution to this problem is RZ which uses 3 values. +V, -V and 0



Each time period is devided into 2 halves
The first half retained at the specific level and return to 0 in the next half.

This scheme requires 2 signal change to encode one bit. which occupies larger bandwidth.

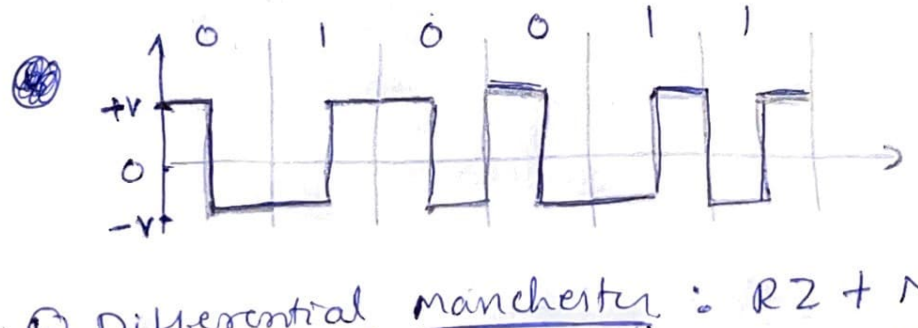
The problems encountered by polar NRZ-L, NRZ-I and RZ are overcomed by Biphase scheme.

Polar Biphase scheme :-

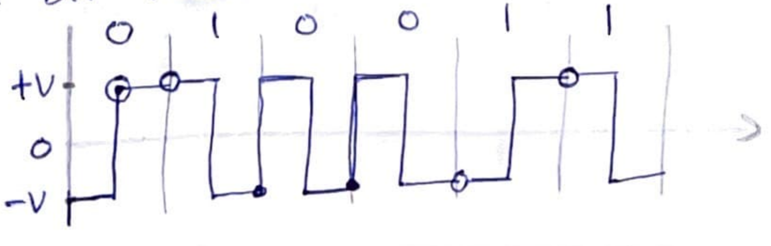
There are two variants.

ⓐ Manchester: Combines RZ + NRZ-L

A bit is devided into 2 halves
voltage remain at one level during first half and moves to other level in second half. This transition at middle of bit provides synchronization.



ⓑ Differential Manchester : RZ + NRZ-I
There is always transition at the middle of bit but the bit values are determined at the begining of bit.



In Manchester and differential Manchester encoding, the transition at the middle of the bit is used for synchronization.

The minimum bandwidth of both these schemes is 2 times that of NRZ.

③ <u>Bipolar Scheme</u>

* This coding sometimes called as multilevel binary.
* Here, three voltage levels are used : +V, -V, and 0.
* The voltage level for one data ele is at 0, while the voltage level for the other ele alternates between +V and -V.

<u>AMI and Pseudoternary</u>

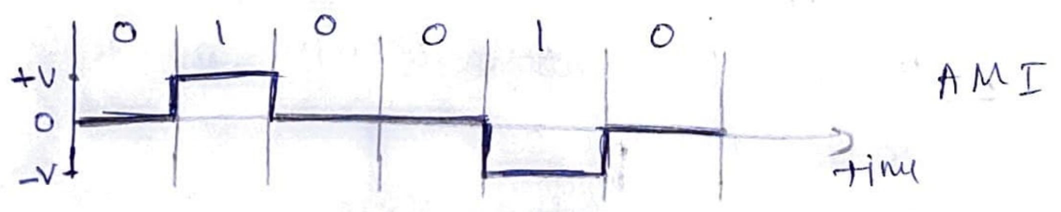
AMI [Alternate Mark Inversion]

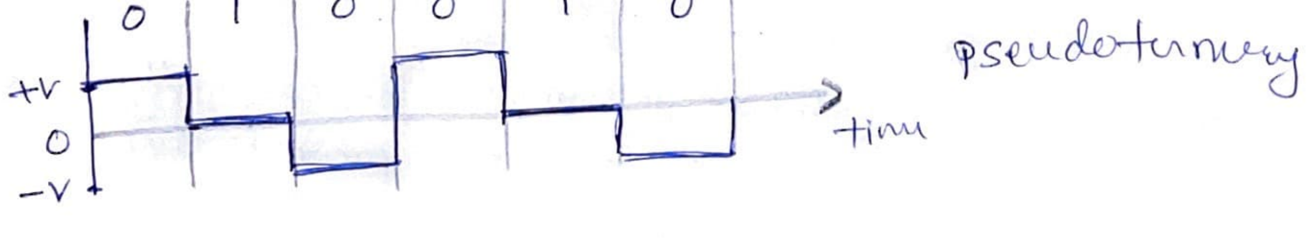
The term "mark" derived from telegraphy meaning 1.

So AMI is Alternate 1 Inverse.

In case of AMI neutral voltage is 0 voltage represents 0 bit while 1's are represented with alternative +V and -V

In case of pseudoternery 0 voltage represents bit 1. and 0's are presented with alternative +V and -V.





* There is no DC component because of following reasons

① In long sequence of 1's, the voltage levels are alternative bet +V and −V; it is not constant

② For long sequence of 0's, the voltage remains constant, but, the amplitude is zero, which is same as having no DC component

AMI is used in long distance comn, but it has a Synchronization problem when a long sequence of 0's is present in the data.

④ Multilevel Schemes

The desire to ↑ data rate or ↓ the required bandwidth has resulted in the creation of many schemes. The main goal is to ↑ no of bits per baud by encoding a pattern of m data ele into a pattern of n signal ele.

* we have 2 type of data ele (0s & 1s)
* So group of m data ele can produce a combination of 2^m pattern.
* If we have L diff signal levels, then we can produce L^n combinations/patterns of signal.
* If $2^m < L^n$, then data pattern occupy only a subset of signal pattern.
* If $2^m = L^n$, then each data pattern is encoded into one signal pattern.
* The subset of signal pattern can be carefully designed to prevent baseline wandering, to provide synchronization and to detect error that occurred during data transmission

* If $2^m > L^n$, then it cannot be encoded because some of the data pattern cannot be encoded.

* This type of coding is classified as mBnL.
  where m – length of binary pattern
  B – Binary data
  n – length of signal pattern.
  L – no of levels in the signal.

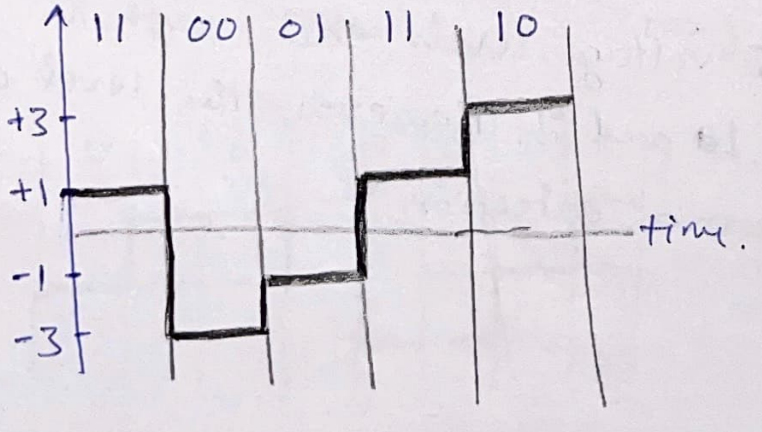
* A letter is used in place of L as follows.
  B = binary ; L = 2
  T = ternery ; L = 3
  Q = quaternery ; L = 4

* In mBnL : mB define data pattern while nL define the signal pattern.

* In mBnL, a pattern of m data ele is encoded as a pattern of n signal ele in which $\underline{2^m \leq L^n}$.

2B1Q (two binary one quantenary) uses data pattern of size 2 & encodes the 2-bit pattern as one signal ele belonging to a 4 level signal

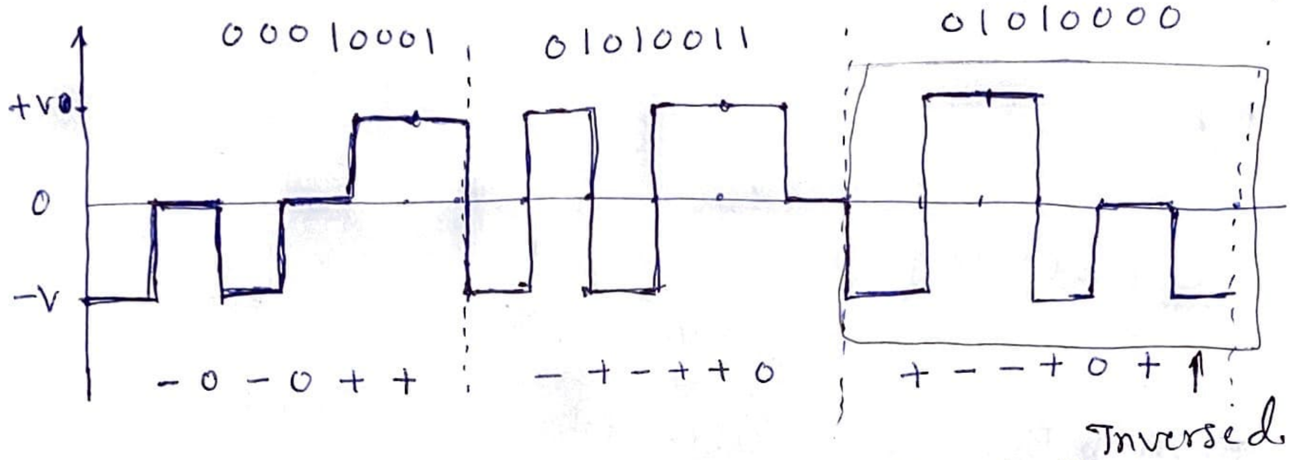
Ex:- Rules : $\boxed{00 \rightarrow -3}$ $\boxed{01 \rightarrow -1}$ $\boxed{10 \rightarrow +3}$ $\boxed{11 \rightarrow +1}$



8B6T (eight binary six ternary)

The idea is to encode a pattern of 8 bits as a pattern of six signal elements, where signal has three levels.

Hence we can have $2^8 = 256$ diff data pattern and $3^6 = 729$ diff signal pattern.



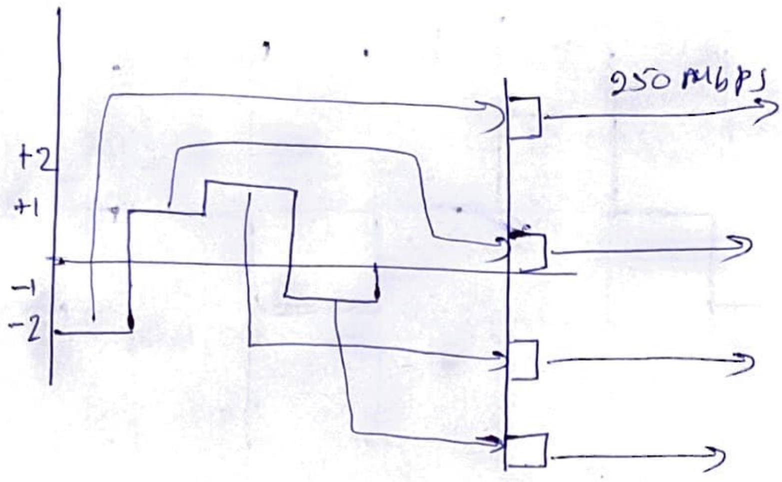
The avg signal rate of the scheme is theoretically

$$S_{avg} = \frac{1}{2} \times N \times \frac{6}{8} \qquad \left[\because S_{avg} = c * N * \frac{1}{r}\right]$$

In practice the minimum bandwidth is very close to $\boxed{6N/8}$.

4D - PAM5 (4-dimensional five level pulse amplitude modulation)

* The data is sent over 4 wires (channels) at a time.
* It uses 5 voltage levels, each with the voltage $-2, -1, 0, 1$ and 2. However, the level 0 is used only for error detection.



Multilevel :-
4D - PAM5
Schem.

(5) <u>Multitransition</u> : MLT -3

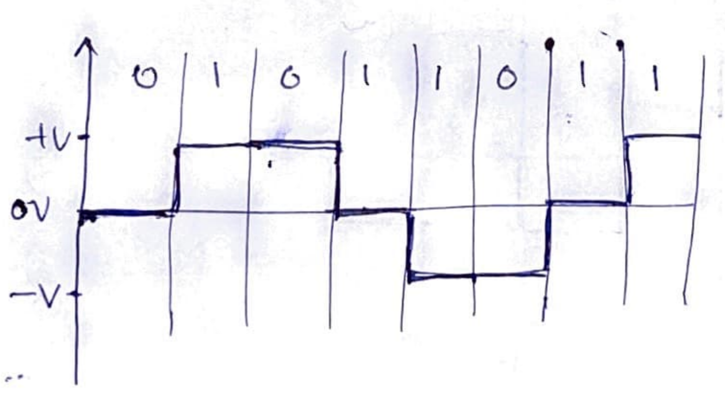
- MLT -3 (Multiline Transition, 3 levels)
uses three level 3 levels (+V, 0 & -V) and 3 transition rules to move bet the levels.

1. If the next bit is 0, there is no transition
2. If the next bit is 1, and the current level is NOT 0, the next level is 0
3. If the next bit is 1, & the current level is 0, the next level is the opposite of the last non-zero level.

i.e. bit 0 → No transition

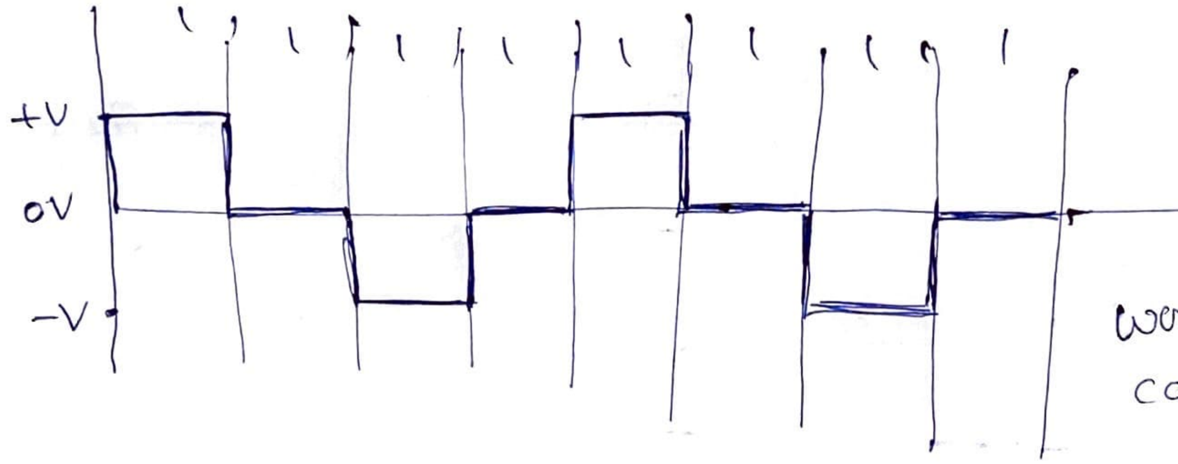
bit 1 + curr level NOT 0 → transition to 0

bit 1 + curr level 0 → transition to opposite of last non-zero level



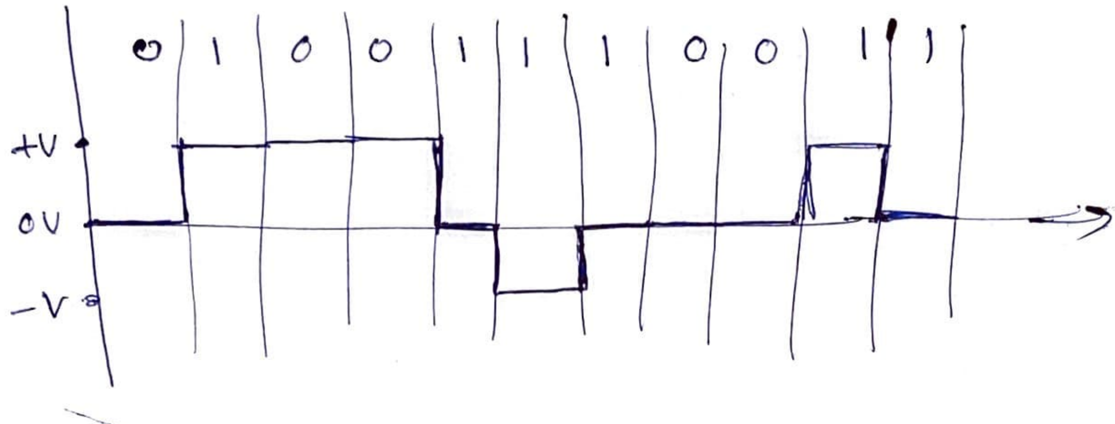
Typical case

(2)



worst case

(3)



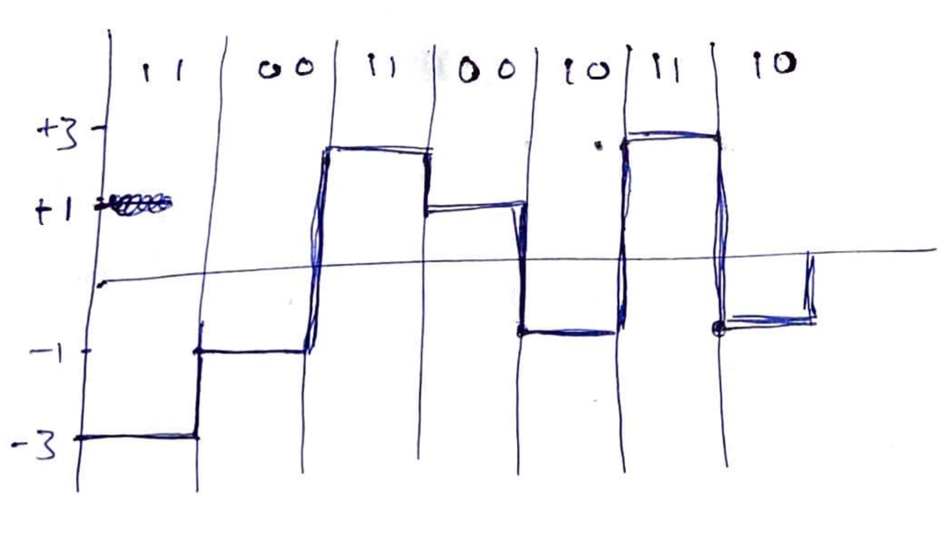
(4) 0 1 1 1 0 1 1 1 0

Exersize for 2B1Q

Digital data : 1100110010111 0

Next bits	prev L→+V Next level	PL→-V Next level
0 0	+1	-1
0 1	+3	-3
1 0	-1	+1
1 1	-3	+3

PL - previous Level



4.2. Analog-to-Digital conversion

Analog signals are ~~the one~~ those created by a microphone or camera.

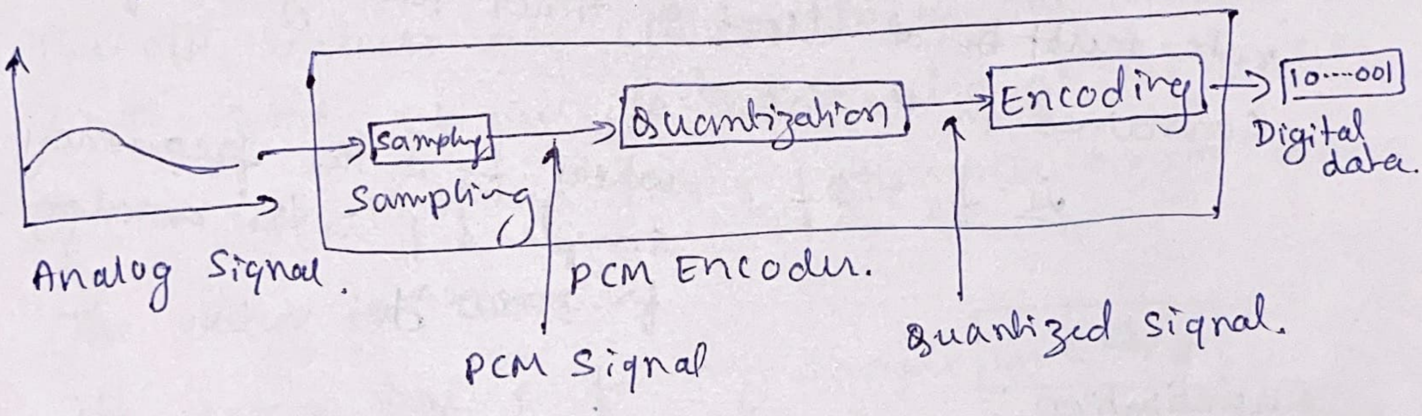
Then ~~the~~ analog signals are to be converted into digital data and then convert to digital signal using any of the techniques studied earlier.

There are two techniques to convert Analog to digital

① PCM ② DM (Delta Modulation)

① PCM (Pulse code modulation)

PCM is common technique used for digitization.
[Digitization means conversion from analog to digital]
PCM encoder has 3 processes as shown in fig



steps:-

① The analog signals are sampled
② The sampled signal is quantized
③ The quantized signals are encoded as stream of bits

① Sampling:-

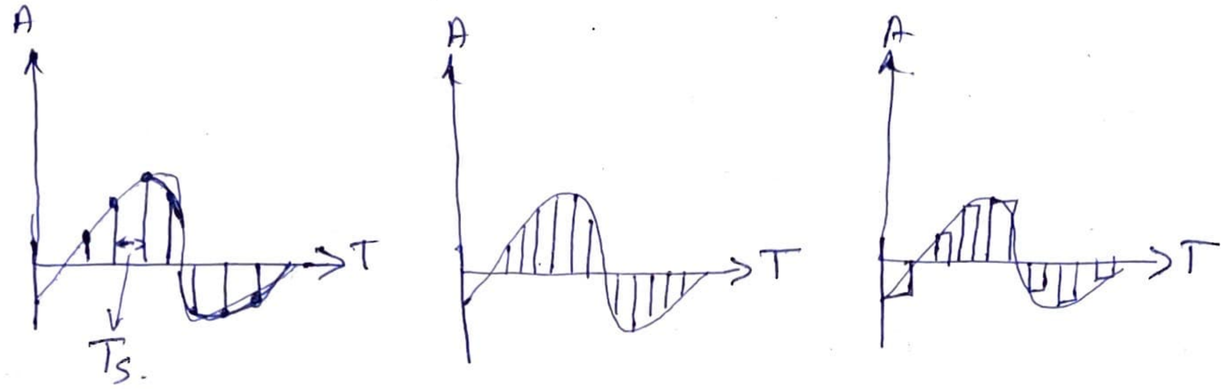
The analog signal sampled every T_s sec. where T_s is the sample interval or period.

The inverse of sampling interval is called the sampling rate or sampling frequency & denoted by f_s. where

$$f_s = \frac{1}{T_s}.$$

There are 3 sampling methods.

① Ideal ② Natural ③ Flat top.



Sampling rate

According to the Nyquist theorem, the sampling rate must be at least 2 times the highest frequency contained in the signal.

i.e. $f_s = 2f$. where f_s is the frequency of sample & f is the analog signal frequency.

Quantization :

The result of sampling is a series of pulses with amplitude values bet the max & min amplitudes of the signal. The steps in quantization

① Assume that the original analog signals has instantaneous amplitudes bet V_{min} & V_{max}.

② Divide the range into L zones each with the height Δ (delta) $\boxed{\Delta = \frac{V_{max} - V_{min}}{L}}$

③ Assign quantized values of 0 ~~to~~ L − 1 to the midpoint of each zone.

④ Approximate the value of the sample amplitude to the quantized values.

Quantized levels

The choice of L, the no of levels, depends on the range of the amplitudes of the analog signal and how accurately we need to recover the signal.

Quantization Error

Quantization is a approximation process, The input is actual signal while o/p is the approximated signal. The o/p values are chosen to be the middle value in the zone. If the i/p value also at the middle of the zone, then no error, otherwise error occurs.

~~The value for error~~

The error value for any sample $\boxed{\text{is} < \Delta/2}$

The range of error is $\boxed{-\frac{\Delta}{2} \le \text{error} \le \frac{\Delta}{2}}$

Uniform versus Nonuniform quantization

In many applications, the distribution of the amplitudes in the analog signal is not uniform.

The changes (amplitude) in lower signal is more often than in high level signals.

For this type of application it is better to use non uniform zones [The Δ is not fixed]

It is greater near the lower amplitudes and less near the higher ~~amp~~ amplitudes

Encoding :-

After each sample is quantized & the no of bits per sample is decided, each sample can be changed to an n_b - bit code word.

The no of bits for each sample is determined from the no of Levels (L). ie $n_b = \log_2 L$.

Ex:- $n_b = \log_2 8$

$n_b = 3$

Bit rate = sampling rate * no of bits/sample

$$\boxed{\text{Bit Rate} = f_s * n_b.}$$

Quantization and encoding of a sampled signal

① Normalized PAM values = $\frac{\text{Actual amplitude}}{\Delta}$ — ①

② Normalized quantized values = Zone mid value. — ②

③ Normalized Error = ① − ②

④ Quantization code = value of the zone band on ②

⑤ Encoded word = Ex quantization code = 2
Encoded word = 010

Refer to text book for diagram/graph

10.3. Cyclic codes

Cyclic codes are special linear block codes with one extra property. In cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword.

Ex:- codeword (a) 1 0 1 1 0 0 0 — one bit cyclic shift → codeword (b) 0 1 1 0 0 0 1

$a_6\ a_5\ a_4\ a_3\ a_2\ a_1\ a_0$ $b_6\ b_5\ b_4\ b_3\ b_2\ b_1\ b_0$

$b_0 = a_6$ $b_1 = a_0$ $b_2 = a_1$ $b_3 = a_2$ $b_4 = a_3$ $b_5 = a_4$ $b_6 = a_5$

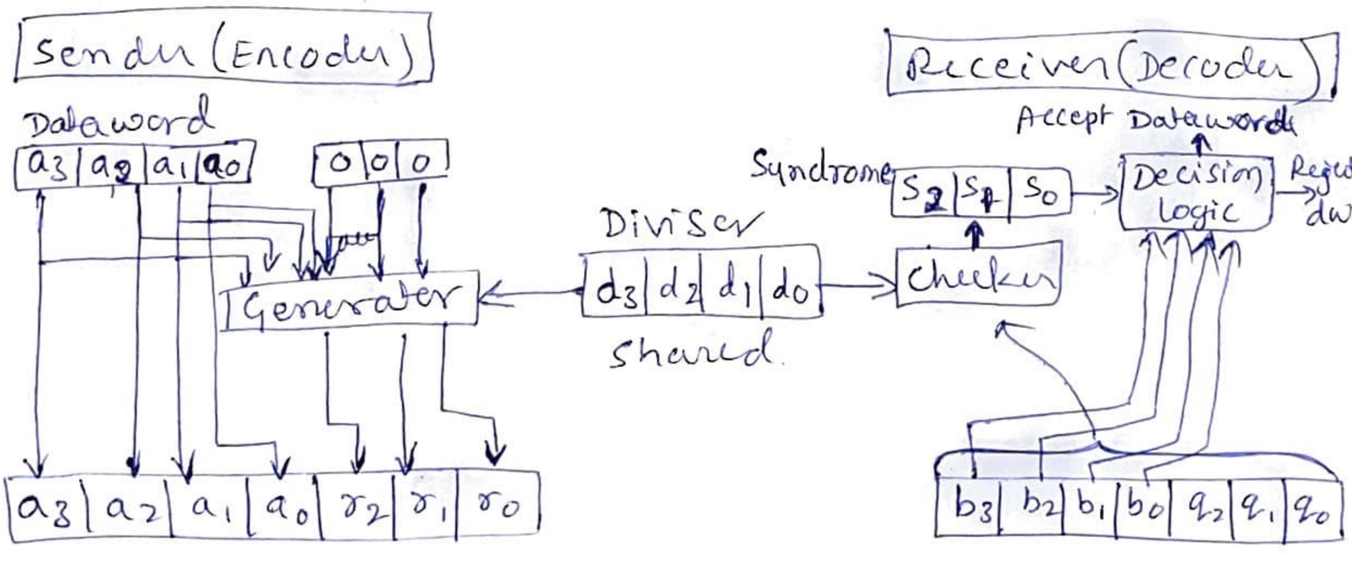
① Cyclic Redundancy check (CRC).

Cyclic codes can be created to correct error.

CRC is a subset of cyclic codes, used in n/w such as LANs and WANs.

Ex:-

Dataword	codeword.
0 0 0 0	0 0 0 0 0 0 0
0 0 0 1	0 0 0 1 0 1 1
0 0 1 0	0 0 1 0 1 1 0
0 0 1 1	0 0 1 1 1 0 1
⋮	⋮
1 1 1 1	1 1 1 1 1 1 1



Description of creating CRC

Dataword – has k bits

codeword – has n bits.

Dataword size is augmented by n-k bits.

Divisor size → n-k+1

Ex :- Encoder for dataword 1001000

Let divisor be 1011

```
          1010
      ___________
 1011 | 1001000
        1011
        -----
        00100
         0000
         -----
         01000
         1011
         -----
          00110
           0000
           -----
           0110   Remainder
```

XOR

a	b	a^b
0	0	0
0	1	1
1	0	1
1	1	0

Generated codeword: dataword 1001 | CRC 110 → This is the codeword sent to the receiver.

Decoder

```
          101
      ___________
 1011 | 1001110
        1011
        -----
        00101
         0000
         -----
         01011
          1011
          -----
          00000 → Syndrome
```

Zero Syndrome

Syndrome will be all zero's if the data is NOT corrupted.

```
          1011
      ___________
 1011 | 1000110
        1011
        -----
        00111
         0000
         -----
         01111
          1011
          -----
          01000
           1011
           -----
           0011
```

Non-zero Syndrome indicates error in the dataword.

② Polynomial :

A pattern of 0's and 1's can be represented as a polynomial with coefficients of 0 & 1.

a_6	a_5	a_4	a_3	a_2	a_1	a_0
1	0	0	0	0	1	1

1 0 0 0 0 1 1

$x^6 + x + 1$

$1x^6 + 0x^5 + 0x^4 + 0x^3 + 0x^2 + 1x^1 + 1x^0$

Binary pattern and corresponding polynomial. ⇒ Short form of the polynomial.

③ cyclic code encoder using polynomial

$$\begin{array}{r|l} & x^3 + x \\ \hline x^3+x+1 & x^6 + \quad\quad \cancel{x^3} \\ & x^6 + x^4 + \cancel{x^3} \\ \hline & \cancel{x^4} \\ & \cancel{x^4} + x^2 + x \\ \hline & \boxed{x^2+x} \end{array}$$

Remainder.

Divide till the degree of remainder is smaller than the degree of divisor

Codeword $\boxed{x^6 + x^3 \mid x^2+x}$

Dataword Remainder.

Media Access Control (MAC)

① CSMA :

To minimize the chance of collision & thereby increase the performance, the CSMA method was developed.

The chance of collision can be reduced if a station senses the medium before trying to use it.

Carrier sense multiple access (CSMA) requires that each station first listen to the ~~do~~ medium before sending

So, CSMA is based on the principle "sense before transmit" or "listen before talk"

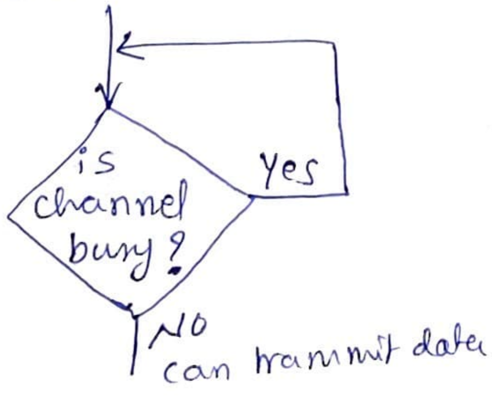
CSMA can reduce the conflict, but cannot eliminate it, as the possibility of collision still exists because of propagation delay.

persistent methods.

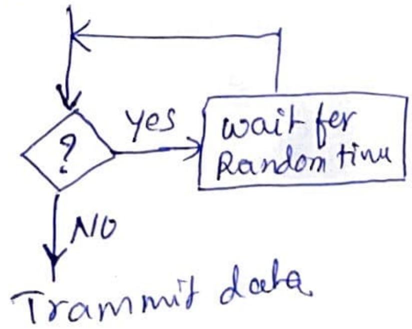
what should a station do if the channel busy?
what should a station do if the channel idle?

To answer these questions, 3 methods have been devised as follows

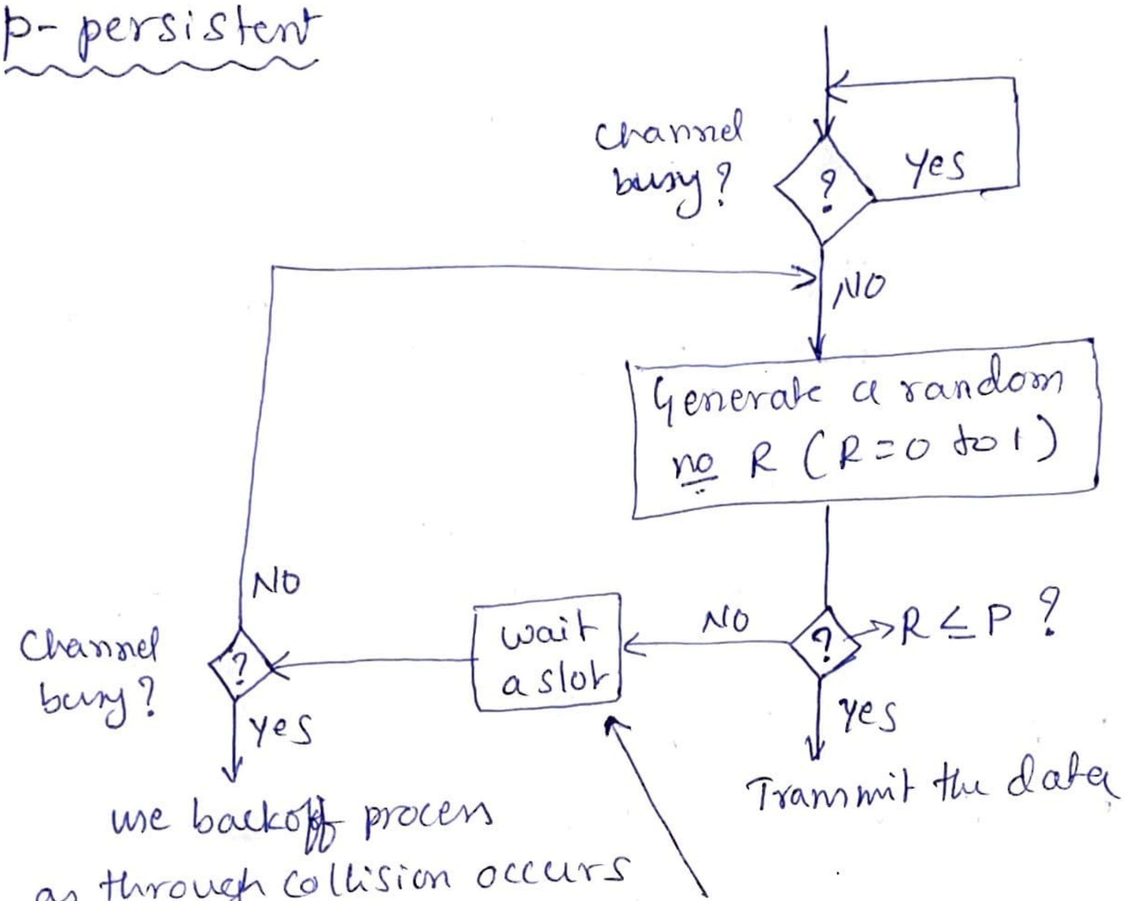
ⓐ 1-persistent



ⓑ Nonpersistent



③ p-persistent



Here the probability of wait is q where $q = 1 - P$.

② CSMA/CD :-

carrier ~~sense~~ sense multiple access with collision detection augments the algorithm to handle the collision.

In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If so, the station is finished. If, however there is a collision, the frame is sent again.

Minimum frame size

For CSMA/CD to work, we need a restriction on the frame size. Before sending the last bit ~~of~~ the frame, the sending station must detect a collision, if any, and abort transmission.

This is because the station, once the entire frame is sent, does not keep a copy of the frame and does not Monitor the line for collision detection.

∴ the frame transmission time T_{fr} must be a least two times the max propogation time T_p.

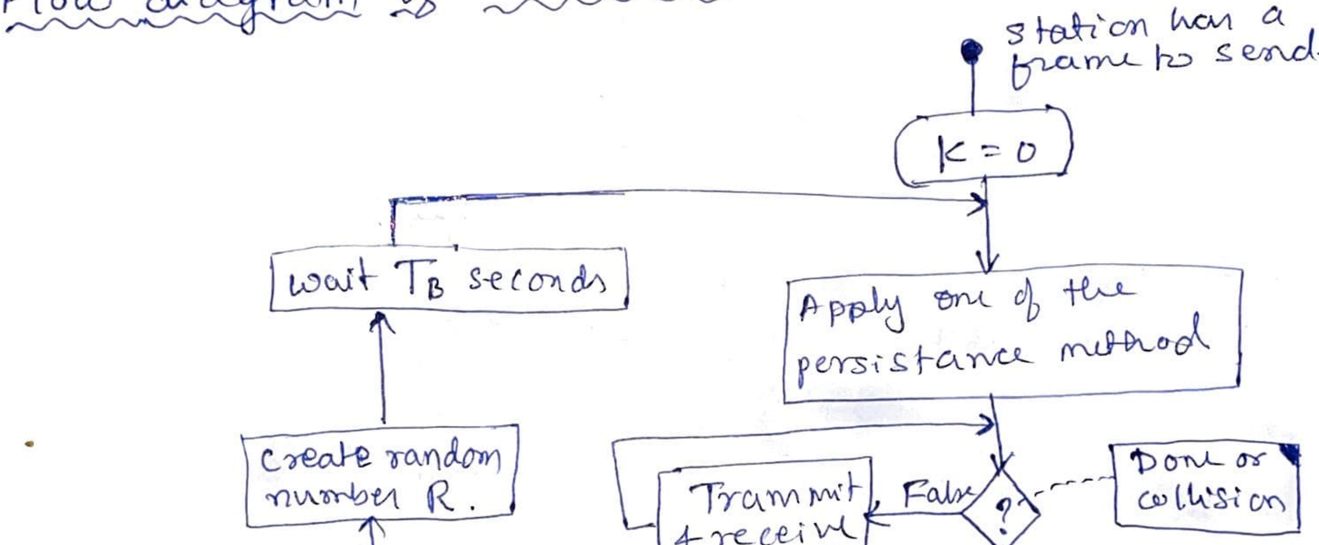
i.e. $T_{fr} = 2T_p$.

$$\frac{L}{Bw} = 2T_p. \qquad \left[\because T_{fr} = \frac{L}{Bw} \right].$$

L is length of frame

Bw is bandwidth.

Flow diagram of CSMA/CD



NOTE :- K : No of attempts

T_{fr} : Frame avg transmission time

R : Random no $(0 \text{ to } 2^K - 1)$

T_B : Back off time $(R * T_{fr})$

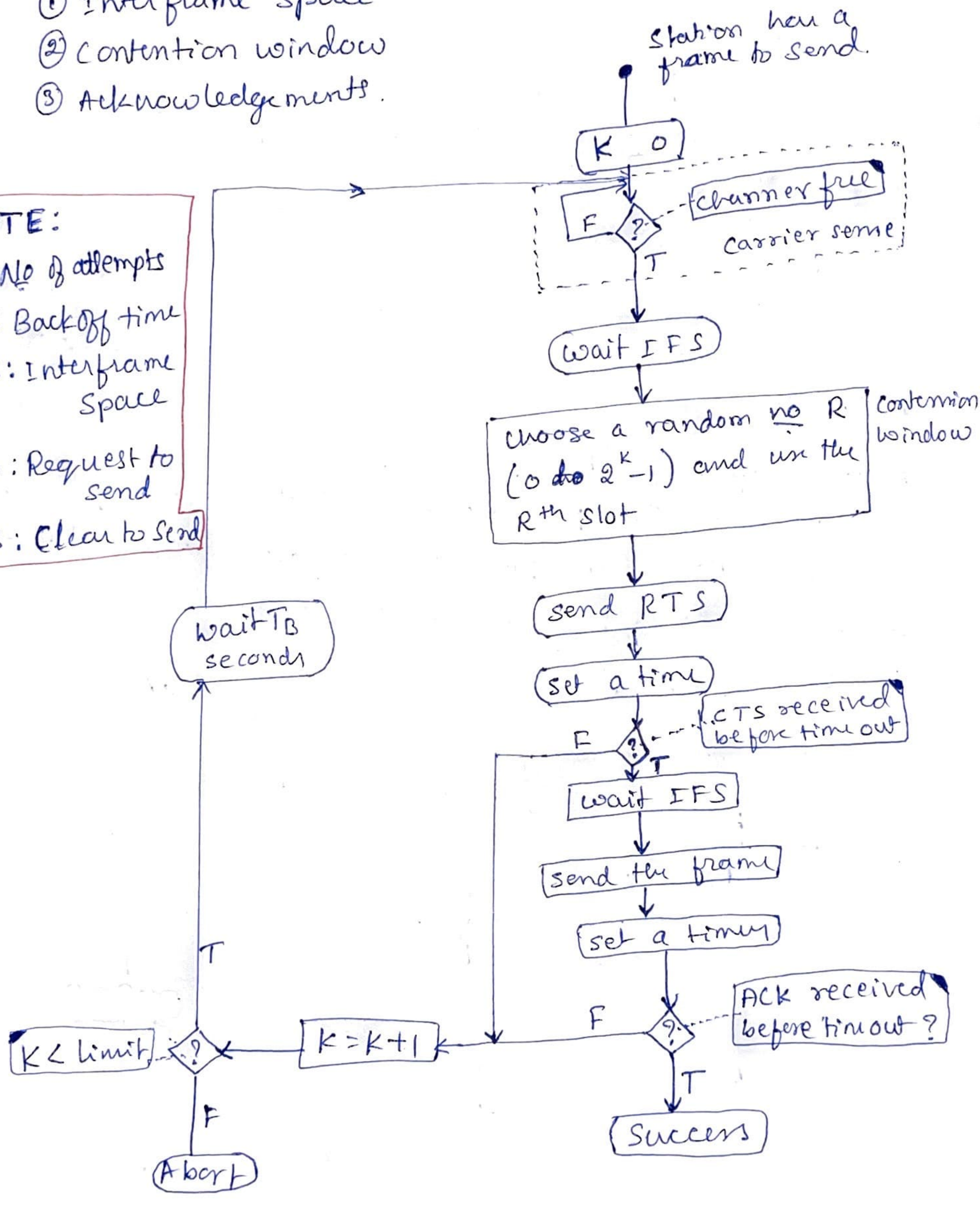
CSMA/CA

Carrier sense multiple access with collision avoidance was invented for wireless n/w. Collisions are avoided with its 3 strategies as shown in fig.

1. Interframe space
2. Contention window
3. Acknowledgements.

NOTE:

K: No of attempts
T_B: Backoff time
IFS: Interframe space
RTS: Request to send
CTS: Clear to send



Interframe space (IFS)

When an idle channel is found, the station instead of sending immediately, waits for a period of time called IFS, thinking that though the channel is empty when it sensed, but a distant station might have already started transmitting.

The IFS variable can be used to prioritize station.

Ex: a station with shorter IFS has higher priority.

Contention window

It is an amount of time divided into slots. A station that is ready to send choose a random no of slots as its wait time.

Acknowledgement (ACK)

With all these precautions still there is a possibility of data getting corrupted, the +ve ack and time-out timers can help to guarantee that the receiver has received the frame properly.

CSMA/CA and NAV flow of data (NAV – N/w Allocation vector)